From Long Term Classroom Programs and Medium / Short Classroom Program 4 in Top 10, 10 in Top 20, 43 in Top 100, 75 in Top 200, 159 in Top 500 Ranks & 3542 total selections in IIT-JEE 2012

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2013

FULL TEST – IV

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 198**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 to 14) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (15 to 20) contains 6 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

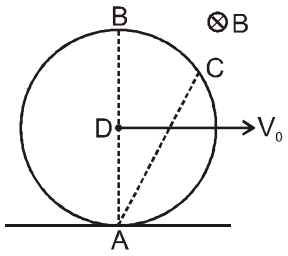
PART – I

SECTION – A

Single Correct Answer Type

This section contain **8 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

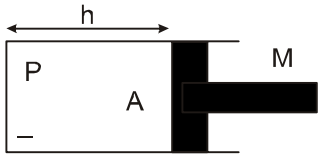
1. A conducting disc of radius R is rolling without sliding on a horizontal surface with a constant velocity of centre of mass, V_0. A uniform magnetic field of strength B is applied normal to the plane of the disc. Then the EMF induced between points B and C will be : (Take AB = 2R, D as centre and length AC = $\sqrt{3}R$).



(A) $\frac{3BV_0R}{2}$ (B) $\frac{5BV_0R}{2}$

(C) $\frac{BV_0R}{2}$ (D) $2BV_0R$

2. A cylindrical conducting piston of mass M slides smoothly inside a long cylinder closed at one end, enclosing a certain mass of a gas. The cylinder is kept with its axis horizontal. If the piston is disturbed from its equilibrium position, it oscillates simple harmonically. The period of oscillation will be (process is isothermal)

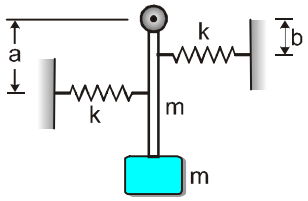


(A) $T = 2\pi\sqrt{\frac{Mh}{PA}}$ (B) $T = 2\pi\sqrt{\frac{MA}{Ph}}$

(C) $T = 2\pi\sqrt{\frac{M}{PAh}}$ (D) $T = 2\pi\sqrt{MPhA}$

Rough work

3. A rod of mass m and length L is hinged at its one end and carries a block of same mass m at its lower end. Two springs of same force constant k are installed at distances a = L/2 and b = L/4 respectively as shown in the figure. If the whole arrangement lies in the vertical plane, the frequency of small vibration is :



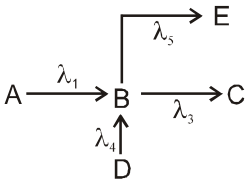
(A) $\frac{1}{2\pi}\sqrt{\frac{\frac{3mgL}{2}+\frac{5kL^2}{4}}{4mL^2}}$

(B) $\frac{1}{2\pi}\sqrt{\frac{3mgL+\frac{5kL^2}{4}}{\frac{4mL^2}{3}}}$

(C) $\frac{1}{2\pi}\sqrt{\frac{\frac{mgL}{2}+\frac{5kL^2}{16}}{\frac{mL^2}{3}}}$

(D) $\frac{1}{2\pi}\sqrt{\frac{\frac{3mgL}{2}+\frac{5kL^2}{16}}{\frac{4mL^2}{3}}}$

4. In radioactive chain reaction shown, initially at t = 0; only N_0 molecules of A are present. At some time t = t, number of molecules of A, B, C, D and E are N_1, N_2, N_3, N_4 and N_5 respectively. Then the activity of B at this time t is :



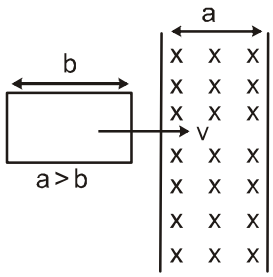
(A) $\lambda_1N_1 + \lambda_4N_4 - \lambda_5N_2 - \lambda_3N_2$

(B) $(\lambda_3 + \lambda_5)\, N_2$

(C) $\lambda_1N_1 + \lambda_4N_4 - \lambda_5N_5 - \lambda_3N_3$

(D) $(\lambda_1 + \lambda_4 - \lambda_3 - \lambda_5)N_2$

5. In the given arrangement, the loop is moved with constant velocity v in a uniform magnetic field B in a restricted region of width a. The time for which the emf is induced in the circuit is:



(A) $\frac{2b}{v}$

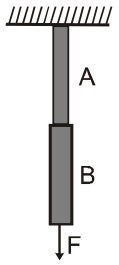
(B) $\frac{2a}{v}$

(C) $\frac{(a+b)}{v}$

(D) $\frac{2(a-b)}{v}$

Rough work

6. The wires A and B shown in the figure, are made of the same material and have unequal radii of cross-section r_A and r_B. When a constant force F is applied as shown, one of the wires breaks.



(A) A will break before B if $r_A > r_B$
(B) B will break before A if $r_A < r_B$
(C) Either A or B could break if $r_A = r_B$
(D) The lengths of A and B must be known to decide which wire will break

7. Two moving lumps of clay collide head on and stick together, the composite lump moving with a kinetic energy which is 80% less than the initial kinetic energy of the two initial lumps. If the final velocity of the lumps is 0.5 m/s, the relative velocity of the lumps before collision is
(A) at least 2 m/s (B) at most 2 m/s
(C) at least 1 m/s (D) at most 1 m/s

8. A uniform magnetic field $\vec{B} = B_0\hat{j}$ exists in space. A particle of mass *m* and charge *q* is projected in the negative *x*-direction with speed *v* from a point (*d*, 0, 0). The maximum value of *v* for which the particle does not hit the yz-plane is
(A) $\frac{Bqd}{m}$ (B) $\frac{2Bqd}{m}$
(C) $\frac{Bqd}{2m}$ (D) $\frac{Bqd}{3m}$

Rough work

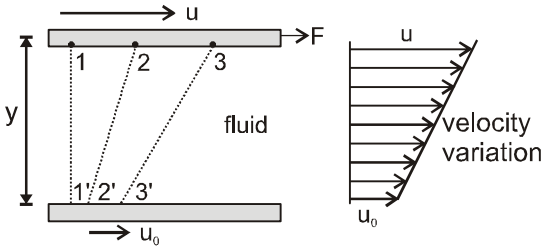
Comprehension Type

This section contains 3 groups of questions. Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Questions 9 and 10

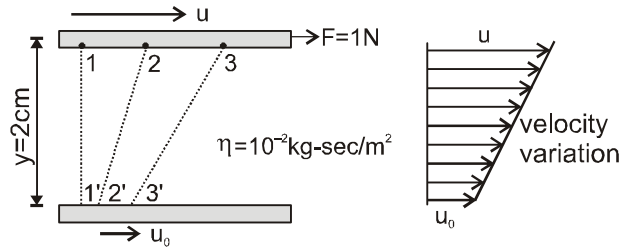
In the shown figure, both the plates are moving. The fluid between the plates moves such that fluid particles in contact of the plates move with same velocity as the plates. The fluid particles in the gap moves in a linear velocity distribution function. As per Newton law of viscosity, the tangential force F is given by

$$F = \eta A \frac{u - u_0}{y}$$



Where η = coefficient of viscosity = 10^{-2} kg -sec/m^3, A = Area of the plate = 1 m^2, y = distance between both the plates = 2 cm. The velocity gradient is given by $\frac{du}{dy}$.

9. If the tangential force F = 1 Newton and u_0 = 1 m/s, then the velocity profile for the given figure is
   (A) u = 100 y
   (B) u = 200 y
   (C) u = 200 y + 1
   (D) u = 100 y + 1

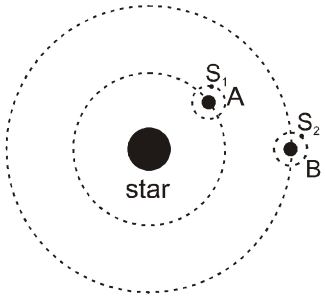


10. In the previous question, the constant velocity of the upper plate would be :
    (A) 1 m/s (B) 2 m/s
    (C) 3 m/s (D) 4 m/s

Rough work

Paragraph for Questions 11 and 12

Consider a star and two planet system. The star has mass M. The planet A and B have the same mass m, radius a and they revolve around the star in circular orbits of radius r and 4r respectively (M > > m, r >> a). Planet A has intelligent life, and the people of this planet have achieved a very high degree of technological advance. They wish to shift a geostationary satellite of their own planet to a geostationary orbit of planet B. They achieve this through a series of high precision maneuvers in which the satellite never has to apply brakes and not a single joule of energy is wasted. S_1 is a geostationary satellite of planet A and S_2 is a geostationary satellite of planet B. Neglect interaction between A and B, S_1 and S_2, S_1 and B & S_2 and A.



11. The ratio of time of revolution of planet A and planet B about star is :

(A) $\frac{T_A}{T_B} = \frac{1}{8}$ (B) $\frac{T_A}{T_B} = 8$

(C) $\frac{T_A}{T_B} = 4$ (D) $\frac{T_A}{T_B} = \frac{1}{4}$

12. If the radius of the geostationary orbit in planet A is given by $r_G = r\left(\frac{m}{M}\right)^{1/3}$, then the time in which the geostationary satellite will complete 1 revolution is

I. 1 planet year = time in which planet revolves around the star

II. 1 planet day = time in which planet revolves about its axis.

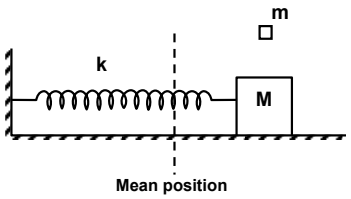
(A) I (B) II

(C) both I and II (D) neither I nor II.

Rough work

Paragraph for Questions 13 and 14

A horizontal spring block system executes SHM with amplitude A = 10 cm initial phase $\phi = 0$ and angular frequency ω. The mass of block is M = 13 kg and there is no friction between the block and the horizontal surface. The spring constant being 2500 N/m.



At $t = t_1$ sec [for which $\omega t_1 = \phi_1 = 30°$]. A mass m = 12 kg is gently put on the block. [Assume that collision between the block and the mass is perfectly inelastic and mass m remains stationary w.r.t. the block M always]

Read above passage carefully and answer the following questions.

13. The new angular frequency of the system will be
    (A) 10 rad/sec (B) 15 rad/sec
    (C) 20 rad/sec (D) none

14. The total energy of system after collision at any moment of time is
    (A) 4 joule (B) 8 joule
    (C) 12 joule (D) none of these

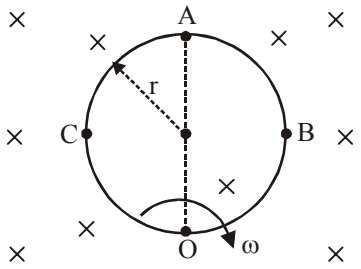
Rough work

Multiple Correct Answer(s) Type

The section contains **6 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct.**

15. A thin paper of thickness 0.02 mm having refractive index 1.45 is pasted across one of the slits in a Young's double slit experiment. The paper transmits 4/9 of light falling on it.
(λ_{light} = 600 nm).
(A) Amplitude of light wave transmitted through the paper will be 2/3 times that of incident wave.
(B) The ratio of maximum and minimum intensity in the fringe pattern will be 25.
(C) The total number of fringe crossing the centre if an identical paper is pasted on the other slit is 15.
(D) The ratio of maximum and minimum intensity in the pattern will be 5.

16. In the figure, there is a conducting ring having resistance R placed in the plane of the paper in a uniform magnetic field B_0. The ring is rotating (in the plane of paper) about an axis passing through point O with constant angular speed ω in clockwise direction.



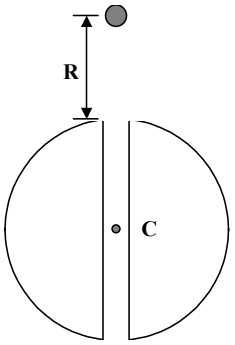
(A) Point A will be at higher potential than O.
(B) The potential of point B and C will be same.
(C) The current in ring will be zero.
(D) The current in the ring will be $\frac{2B_0\omega.r^2}{R}$.

17. Which of the following will have a different time period (compared to the earth), if taken to the moon?
(A) A simple pendulum.
(B) A spring mass system oscillating vertically in the gravitational field.
(C) A torsion pendulum.
(D) An hourglass clock, which is essentially a vessel filled with a liquid, emptying through a hole in the bottom.

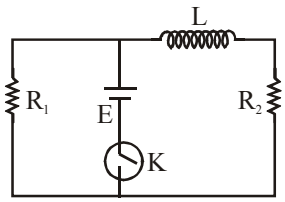
Rough work

18. Hydrogen atoms absorbs radiation of wavelength λ_0 and consequently emit radiations of 6 different wavelengths of which three wavelengths are shorter than λ_0.
(A) The final excited state of the atoms is n = 4
(B) The initial state of the atoms may be n = 2
(C) The initial state of the atoms may be n = 3
(D) There are three transitions belonging to Lyman series.

19. A particle is dropped from a height equal to the radius of the earth above the tunnel dug through the earth as shown in the figure.
R : Radius of earth.
M : Mass of earth.
(A) Particle will oscillate through the earth to a height R on both sides
(B) Particle will execute simple harmonic motion
(C) Motion of the particle is periodic
(D) Particle passes the centre of earth with a speed = $\sqrt{\frac{2GM}{R}}$



20. In the L–R circuit shown in the figure, the key K is closed at time t = 0.
(A) the current through R_1 decreases with time, t
(B) the drop across L increases with t
(C) the magnetic energy stored, in the steady state, equals $\frac{1}{2}L\frac{E^2}{R_2^2}$
(D) the total current through the battery is, in the steady state, $\frac{E}{R_1}+\frac{E}{R_2}$



Rough work

Chemistry

PART – II

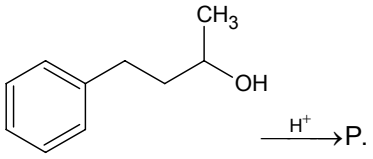
SECTION – A

Straight Objective Type

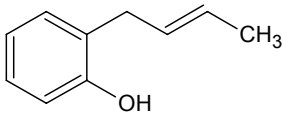
This section contains 8 multiple choice questions numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

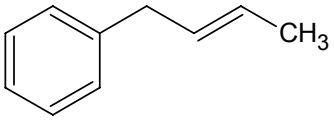
1. In an amino acid the carboxylic acid group has $K_a=10^{-4}$ and amino group has $K_b= 10^{-5}$. The isoelectric point of that amino acid is
   (A) 4 (B) 5
   (C) 6.5 (D) 4.5

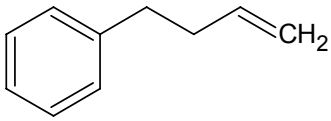
2. Why do mercury batteries and lead acid batteries maintain approximate constant voltage?
   (A) They never run down
   (B) No aqueous species appear in the equation for the overall cell reaction
   (C) Their E° values are very high
   (D) The concentration of reagents in the half cell reactions remain approximately constant

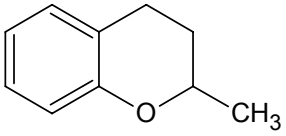
3. 

   The product P is

   (A) 

   (B) 

   (C) 

   (D) 

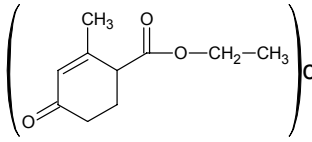
4. Which of the following quantities is proportional to the electron density at a point?
   (A) the wave function (B) the de-Broglie wavelength
   (C) the square of wave function (D) the reciprocal of the de Broglie wavelength

Rough Work

5. Which of the following will give maximum number of isomers?

(A) $\left[Co(NH_3)_4Cl_2\right]$ (B) $\left[Ni(en)(NH_3)_4\right]^{2+}$

(C) $\left[Ni(C_2O_4)(en)_2\right]$ (D) $\left[Cr(SCN)_2(NH_3)_4\right]^{2+}$

6. A compound Hagemann's Ester  can be prepared by treating a mixture of formaldehyde and ethylacetoacetate. Following steps are used

1. Itnramolecular aldol condensation.
2. Acid hydrolysis followed by decarboxylation
3. Michael addition
4. Aldol like condensation

What is the correct sequence of reactions:

(A) $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ (B) $4 \rightarrow 3 \rightarrow 1 \rightarrow 2$

(C) $4 \rightarrow 3 \rightarrow 2 \rightarrow 1$ (D) $2 \rightarrow 3 \rightarrow 4 \rightarrow 1$

7. Which of the following compounds on decomposition, releases two gases containing the same elements?

(A) $FeSO_4$ (B) $FeCO_3$

(C) $Fe(NO_3)_2$ (D) $Fe(ClO_3)_2$

8. If the energy of H-atom in the ground state is – E, the velocity of photo-electron emitted when a photon having energy E_p striked a stationary Li^{2+} ion in ground state, is given by:

(A) $V = \sqrt{\frac{2(E_p - E)}{m}}$ (B) $V = \sqrt{\frac{2(E_p + 9E)}{m}}$

(C) $V = \sqrt{\frac{2(E_p - 9E)}{m}}$ (D) $V = \sqrt{\frac{2(E_p - 3E)}{m}}$

Rough Work

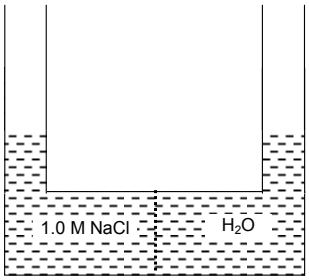
Comprehension Type

This section contains 3 groups of questions. Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 9 to 10

Read the paragraph carefully and answer the following questions:

Osmosis is a process of movement of solvent particles from a region of lower to higher concentration through semi permeable membrane. The pressure exerted by the liquids on semi permeable membrane at the time of equilibrium is called osmotic pressure. If the pressure more than osmotic pressure is applied at the more concentration side, reverse osmosis takes place, which is used for the purification of water



9. Select the correct statement for the above diagram
   (A) If semi permeable membrane is permeable to only H_2O, level of liquid in the left arm will decrease
   (B) If membrane is permeable to H_2O, Cl^- and Na^+ all, the level of liquid in two arms is different
   (C) In the reverse osmosis process water will move from NaCl side to H_2O side
   (D) all of these

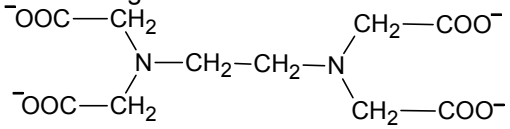
10. A water desalination plant is set up near a salt marsh containing water that is 0.1 NaCl. The minimum pressure (P) that must be applied at 27°C to purify the water by reverse osmosis is
   (A) $P \geq 4.926$ atm (B) $P \geq 2.46$ atm
   (C) $P \leq 2.46$ atm (D) $P \leq 4.926$ atm

Rough Work

Paragraph for Question Nos. 11 to 12

Read the paragraph carefully and answer the following questions:

EDTA is used as a complexing agent in chemical analysis. Solution of EDTA usually containing the disodium salt Na_2H_2 EDTA are also used to treat heavy metal poisoning. EDTA is a hexadentate ligand having the following structure.



Answer the following questions

11. A solution having equal moles of both reactant show the following equilibrium (Keq=10^{-3})

$Pb^{2+}(aq) + H_2\,EDTA^{2-}(aq) \rightleftharpoons Pb\,EDTA^{2-}(aq) + 2H^{+}(aq)$

If concentration of complex at equilibrium is 10^{-3} M find the concentration of Pb^{2+} at equilibrium maintained at pH = 3

(A) 10^{-9} (B) 10^{-6}

(C) 10^{-3} (D) None of these

12. EDTA forms stable complexes with many metal ions due to

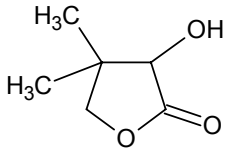
(A) High negative charge density (B) Chelation

(C) Low negative charge density (D) Because it is very weak field ligand

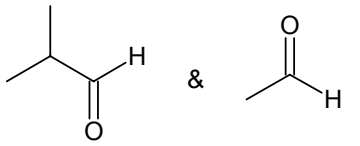
Rough Work

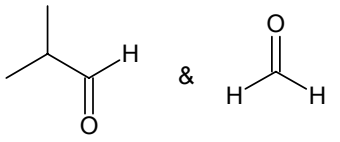
Paragraph for Question Nos. 13 to 14

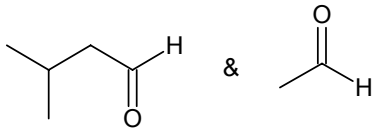
The aliphatic aldehyde 'P' and 'Q' react in the presence of aqueous K_2CO_3 to give compound 'R', which upon treatment with HCN provides compound 'S'. On acidification and heating, 'S' gives the product shown below:

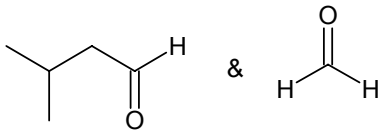


13. The compounds P and Q respectively are

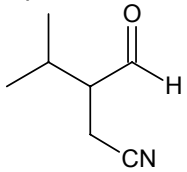
(A) 

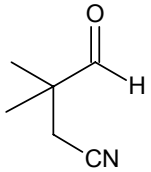
(B) 

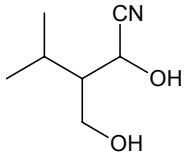
(C) 

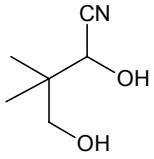
(D) 

14. The compound S is

(A) 

(B) 

(C) 

(D) 

Rough Work

Multiple Correct Choice Type

This section contains 6 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

15. 

Correct statement for product B is
(A) It is gives positive carbyl amine test
(B) It gives coupting reaction with diazonum salt
(C) It gives alcohol on treatment with nitrous acid
(D) It forms dibromo derivative with bromine water

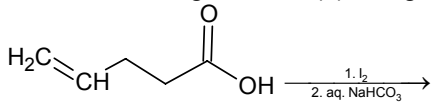
16. Trans-2-butene $\xrightarrow{CF_3CO_3H} A \xrightarrow{H_3O^+} B \xrightarrow{HIO_4} C$

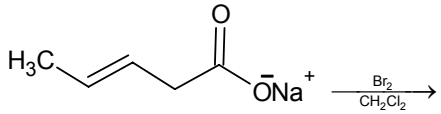
The incorrect statement is
(A) The product B is optically active
(B) Product C gives the iodoform test
(C) Product C does not gives iodoform test
(D) Product A is an alcohol

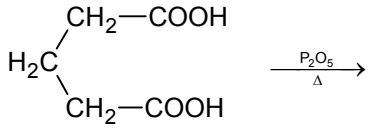
17. For which of the following species it is possible to solve equation analytically and obtain an equation for its electronic energy level
(A) He (B) Li^{2+}
(C) H_2 (D) H

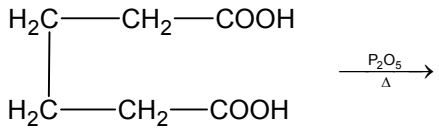
Rough Work

18. LiF and NaBr are ionic salts that both form a face centred cubic lattice. The ionic radii of Li^+, F^-, Na^+ and Br^- are 0.76, 1.33, 1.02 and 1.96 A^0 respectively, which of the following statement is / are true
(A) The atomic radius of Na is smaller than 1.0 A^0
(B) The atomic radius of F is smaller than 1.33 A^0
(C) Each Li^+ ion in LiF has $8F^-$ ions as nearest neighbors
(D) The lattice energy of LiF is greater in magnitude than the lattice energy of NaBr

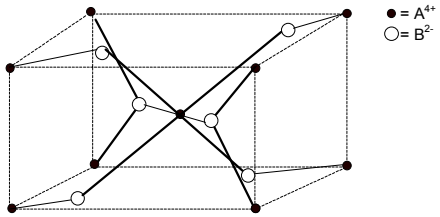
19. Which of the following reaction (s) can generate a five membered ring as the major product?
(A) 

(B) 

(C) 

(D) 

20. Consider the following solid:



(distarted cubic structure)

Identify the correct statements regarding the above crystalline solid:
(A) The coordination number of A^{4+} is 8
(B) The coordination number of B^{2-} is 3
(C) The formula of the solid is AB_2
(D) Effective number of A^{2+} ions per unit cell is 2

Rough Work

Mathematics

PART – III

SECTION – A

Single Correct Choice Type

This section contains 8 multiple choice questions numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Consider P: $y^2 = 4ax$ and RH: $x^2 - y^2 = a^2$. Let S = {m∈R/m is the slope of the normal to P that touches the RH}. Then |S| is
   (A) 0 (B) 2
   (C) 4 (D) 6

2. Let x, y, z∈R. The number of integral values that x can take if x + y +z = 5 and $x^2 + y^2 + z^2 = 9$, is
   (A) 0 (B) 1
   (C) 2 (D) infinitely many

3. The number of solutions of the equations $2\sin^2 x + \sin^2 2x = 2$, $\sin 2x + \cos 2x = \tan x$ in $[0, 4\pi]$ satisfying the condition $2\cos^2 x + \sin x \le 2$ is
   (A) 4 (B) 6
   (C) 8 (D) 2

4. If |z| = 1 and $z \neq 0$, then the points $z + \frac{1}{z}$ lie on a
   (A) straight line not through the origin (B) circle
   (C) line segment lying on the x-axis (D) line segment lying on the y-axis

5. The vectors $\left(x^2-1\right)\bar{i} + 2\left(x^2-1\right)\bar{j} - 3\left(x^2-1\right)\bar{k}, \left(2x^2-1\right)\bar{i} + \left(2x^2+1\right)\bar{j} + x^2\bar{k}$, and $\left(3x^2+2\right)\bar{i} + \left(x^2+4\right)\bar{j} + \left(x^2+1\right)\bar{k}$ are non-coplanar. The number of real values that x cannot take is
   (A) 0 (B) 1
   (C) 2 (D) 4

Rough work

6. Let $f(x)=\left(a-x^n\right)^{1/n}$ and $g(x)=\underbrace{f\circ f\circ f\circ.....\circ f(x)}_{2m \text{ times}}$. Then $\int \frac{dx}{(g(x))^n\left(1+(g(x))^n\right)^{1/n}}$ equals

(A) $\frac{n}{n-1}\left(1+\frac{1}{x^n}\right)^{\frac{n-1}{n}}+C$

(B) $\frac{1}{1-n}\left(1+x^{-n}\right)^{\frac{n-1}{n}}+C$

(C) $\frac{1}{1+n}\left(1+x^n\right)^{\frac{n+1}{n}}+C$

(D) $\frac{-1}{n}\left(1-x^n\right)^{\frac{n}{n+1}}+C$

7. If $y=f(x)$ and $z=g(x)$, then $\frac{d^2y}{dz^2}$ equals

(A) $\frac{g'f''-f'g''}{(g')^2}$

(B) $\frac{g'f''-f'g''}{(g')^3}$

(C) $\frac{g'f'-f''g''}{(g')^3}$

(D) $\frac{g'f''+f'g''}{(g')^3}$

8. The function f(x) = $\begin{cases} \frac{a\,|x^2-x-6|}{6+x-x^2}, x<3 \\ b\lim\limits_{x\to 3}\left(\frac{\int_3^x t\,dt}{x-3}\right), x=3 \\ \frac{x-[x]}{x-3}, x>3 \end{cases}$

is continuous at x = 3. If f (3) $\neq$ 0, then {a, b} is (where [ ] denotes the greatest integer function)

(A) {3, 1}

(B) $\left\{\frac{1}{3},1\right\}$

(C) $\left\{1,\frac{1}{3}\right\}$

(D) $\left\{1,\frac{1}{9}\right\}$

Rough work

Comprehension Type

This section contains **3 paragraphs**. Based upon paragraph **2 multiple choice questions** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of **which ONLY ONE** is correct.

Paragraph for Question Nos. 9 to 10

Read the following write up carefully and answer the following questions:

Let S_r, $r \in N$, denote the sum of an infinite GP whose first term is $2r - 1$ and common ratio $\frac{1}{2r.}$. Let $V_n = \sum_{r=1}^{n} S_r$ and $Q_n = \sum_{r=1}^{n} \frac{1}{S_r S_{r+1}}$.

9. $\lim_{n\to\infty} \frac{V_n}{n^2}$ equals

(A) 0 (B) -1
(C) 1 (D) 2

10. Consider the following statements:
I : $Q_1 > Q_2 > Q_3 > \ldots\ldots$
II: $V_1 < V_2 < V_3 < \ldots\ldots$
Then
(A) Only I is true (B) Only II is true
(C) Both I and II are true (D) Neither I nor II is true

Paragraph for Question Nos. 11 to 12

Read the following write up carefully and answer the following questions:

Consider the function $f : R \to R$ defined by $f(x) = a^{-x} + a^{|x|},\ a > 1$

11. The interval in which f(x) is monotonic decreasing, is

(A) (0, a) (B) $(-\infty, 0)$
(C) $(0, \infty)$ (D) (2, 2a)

Rough work

12. Area bounded by the given curve and the lines $x = -1$, $x = 1$ and $y = 0$, is

(A) $\dfrac{2a^2+3a+1}{\log_e a}$ (B) $\dfrac{3a^2-2a-1}{a\log_e a}$

(C) $\dfrac{3a-1}{a\log_e a}$ (D) $\dfrac{(3a-1)(2a-1)}{\log_e a}$

Paragraph for Question Nos. 13 to 14

Read the following write up carefully and answer the following questions:

Let S = set of triplets (A, B, C) where A, B, C are subsets of {1, 2, 3, ... n}. E_1 = event that a selected triplet at random from set S will satisfy $A\cap B\cap C = \phi$, $A \cap B \neq \phi$, $B \cap C \neq \phi$. E_2 = events that a selected triplet at random from set S will satisfy $A\cap B\cap C = \phi$, $A\cap B \neq \phi$ $B\cap C \neq \phi$, $A\cap C \neq \phi$. P (E) represents probability of an event E than

13. $P(E_1)$ is equal to

(A) $\dfrac{7^n-6^n+5^n}{8^n}$ (B) $\dfrac{7^n-2\times 6^n+5^n}{8^n}$

(C) $\dfrac{7^n-2\times 6^n}{8^n}$ (D) $\dfrac{7^n-2\times 6^n+5^n}{8^n}$

14. $P(E_2)$ is equal to

(A) $\dfrac{7^n-3\times 6^n+5^n}{8^n}$ (B) $\dfrac{7^n-3\times 6^n+3\times 5^n-4^n}{8^n}$

(C) $\dfrac{7^n-2\times 6^n+2\times 5^n-4^n}{8^n}$ (D) $\dfrac{7^n-6^n+5^n-4^n}{8^n}$

Rough work

Multiple Correct Answer(s) Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

15. The straight line $b^{1/3}y + a^{1/3}x + a^{2/3}b^{2/3} = 0$
(A) touches the parabola $y^2 = 4ax$
(B) touches the parabola $x^2 = 4by$
(C) intersects both the parabolas $y^2 = 4ax$ and $x^2 = 4by$ in real points
(D) touches $y^2 = 4ax$ and intersects $x^2 = 4by$

16. If x, y, a, b are real numbers such that $(x + iy)^{1/5} = a + ib$ and $p = \frac{x}{a} - \frac{y}{b}$, then
(A) a – b is a factor of p
(B) a + b is a factor of p
(C) a + ib is a factor of p
(D) a – ib is a factor of p

17. Let $0 < \theta < \frac{\pi}{4}$. Then,
(A) $\sum_{k=0}^{\infty} \sin^{2k}\theta = \sec^2\theta$
(B) $\sum_{k=0}^{\infty} \cos^{2k}\theta = \operatorname{cosec}^2\theta$
(C) $\sum_{k=0}^{\infty} \tan^{2k}\theta = \frac{\cos^2\theta}{\cos 2\theta}$
(D) $\sum_{k=0}^{\infty} \tan^{2k}\theta + \sum_{k=0}^{\infty} \cot^{2k}\theta = 1$

18. A number is chosen at random from the set of integers {1, 2, 3, 4,……., n}. Let A denote the event that the number chosen is divisible by 4, B denote the event that the number chosen is divisible by 5 and C denote the event that the number chosen is divisible by 7. Then,
(A) A, B, C are always mutually independent
(B) A and B are always independent
(C) B and C are independent if n is of the form 35k (k positive integer)
(D) A and C are independent if n is of the form 28 λ (λ positive integer)

Rough work

19. Let A be the set of real values of x, and B be the set of real values of y, satisfying the equation $3^x + 3^y = 9$. Let a = [x], $x \in A$, and b = [y], $y \in B$. Then
(A) The roots of the equation $x^3 + 2x^2 - x - 2 = 0$ are x_1, x_2, x_3 such that $x_1, x_2, x_3 \in A$.
(B) The roots of the equation $x^3 - 2x^2 - x + 2 = 0$ are x_1, x_2, x_3 such that $x_1, x_2, x_3 \in B$.
(C) $a_{max} + b_{max} = 2$
(D) $17a_{max} - 12b_{max} = -5$

20. Let T be the Δ with vertices (0, 0), (0, c^2) and (c, c^2) and let R be the region between y = cx and $y = x^2$ where c > 0 then

(A) Area of $(R) = \frac{c^3}{6}$ (B) Area of $R = \frac{c^2}{3}$

(C) $\lim_{c \to 0^+} \frac{\text{Area}(T)}{\text{Area}(R)} = 3$ (D) $\lim_{c \to 0^+} \frac{\text{Area}(T)}{\text{Area}(R)} = \frac{3}{2}$

Rough work